Distortion-01-BMP

Project Report

1. Readme

Distortion pedal, loosely based on BMP

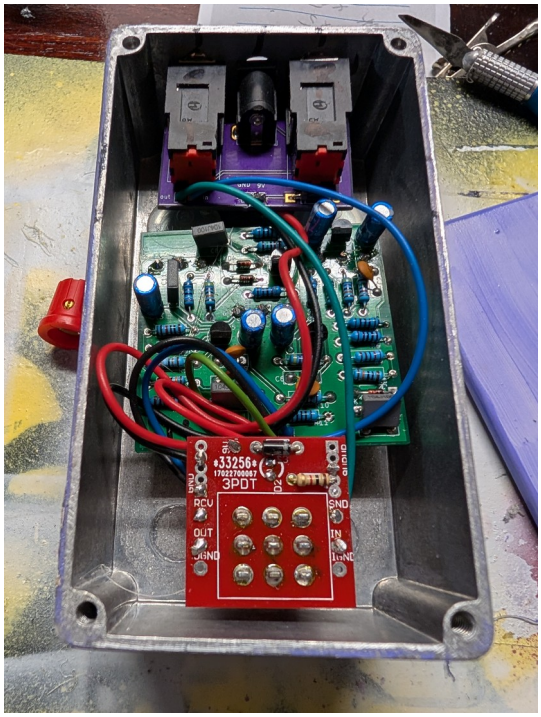
Two PCB layouts now, one for not 100% clear version of BMP, and the other based on the ramshead version. This is boxed up with an old 3pdt pcb, and one of the first tries at boxing with first cut at top-jacks. A little tight. The electolytic caps should not be that close to the jacks pcb.

The latter (ramshead) is in queue to be verified....

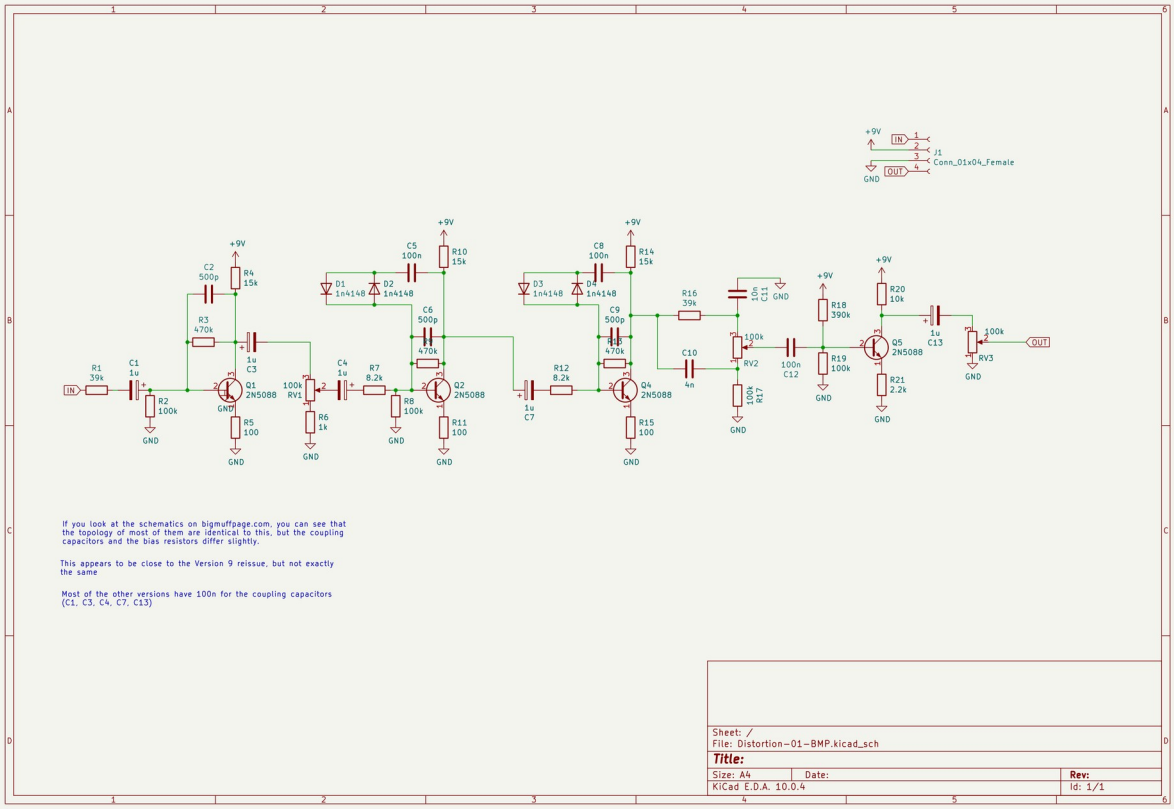
Requires KICAD 7

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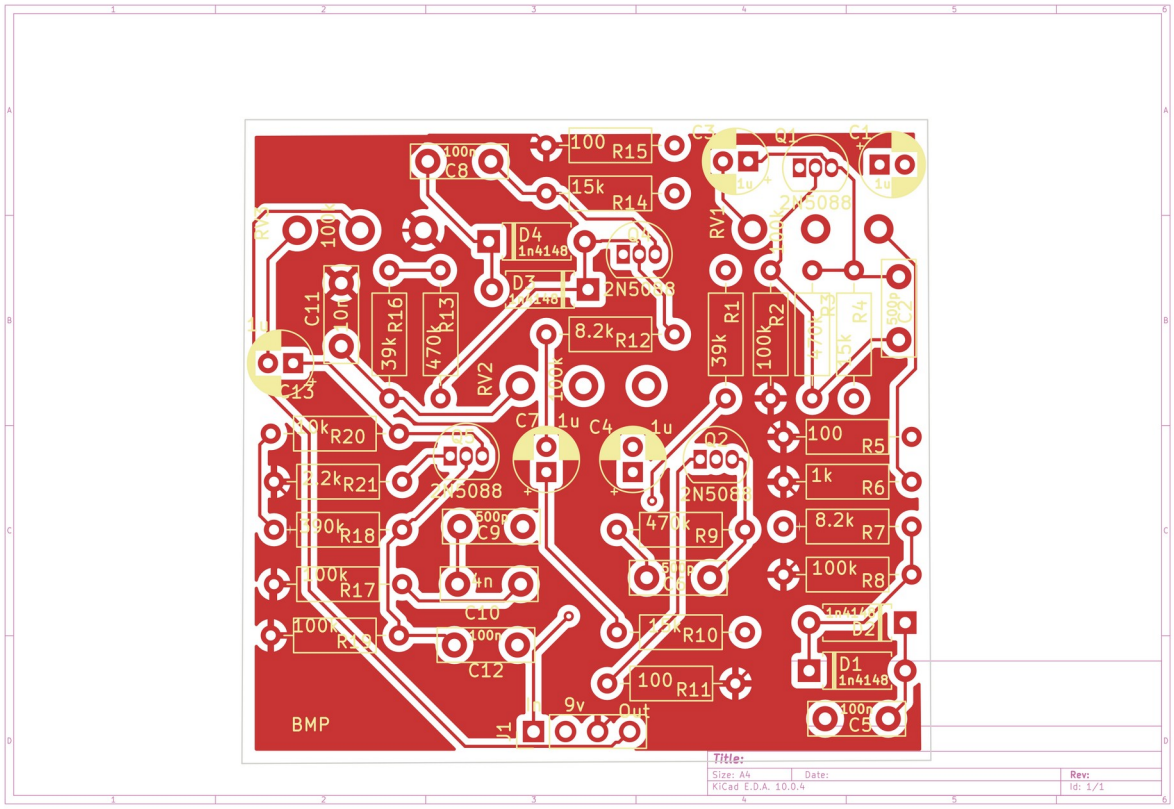




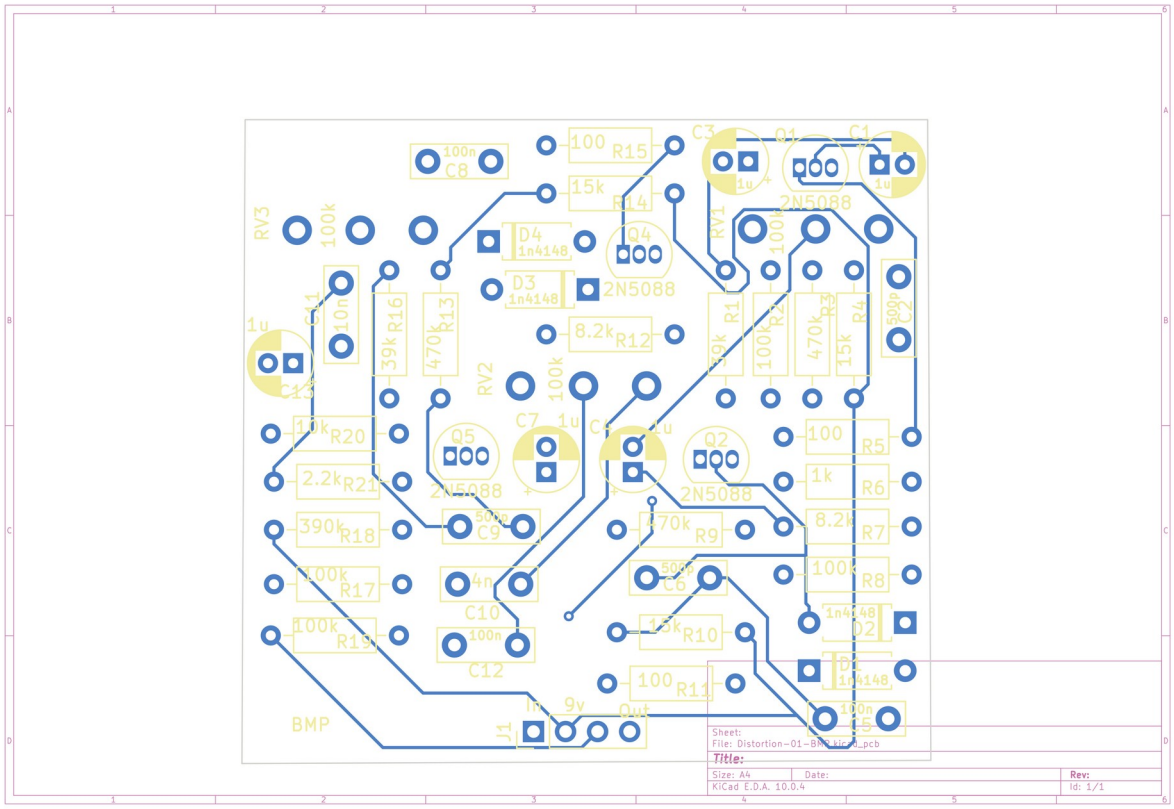
2. Schematic


+9V
IN
OUT
GND
J1
Conn_01x04_Female
R1 39k
C1 1u
R2 100k
C2 500p
R3 470k
R4 15k
C3 1u
Q1 2N5088
R5 100
100k RV1
R6 1k
C4 1u
R7 8.2k
R8 100k
C5 100n
R10 15k
D1 1n4148
D2 1n4148
C6 500p
470k
Q2 2N5088
R11 100
C7 1u
R12 8.2k
D3 1n4148
D4 1n4148
C8 100n
R14 15k
C9 500p
R13 470k
Q4 2N5088
R15 100
R16 39k
C10 4n
C11 10n
100k RV2
470k R17
C12 100n
R18 390k
R19 100k
R20 10k
Q5 2N5088
R21 2.2k
C13 1u
100k RV3
If you look at the schematics on bigmuffpage.com, you can see that the topology of most of them are identical to this, but the coupling capacitors and the bias resistors differ slightly.
This appears to be close to the Version 9 reissue, but not exactly the same
Most of the other versions have 100n for the coupling capacitors (C1, C3, C4, C7, C13)
Sheet: /
File: Distortion-01-BMP.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1

3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



5. Bill of Materials

Reference	Value	Footprint	Qty	DNP
C1,C3,C4,C7,C13	1u	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm	5	
C2,C6,C9	500p	Capacitor_THT:C_Disc_D7.5mm_W2.5mm_P5.00mm	3	
C5,C8,C12	100n	Capacitor_THT:C_Disc_D7.5mm_W2.5mm_P5.00mm	3	
C10	4n	Capacitor_THT:C_Disc_D7.5mm_W2.5mm_P5.00mm	1	
C11	10n	Capacitor_THT:C_Disc_D7.5mm_W2.5mm_P5.00mm	1	
D1-D4	1n4148	Diode_THT:D_A-405_P7.62mm_Horizontal	4	
J1	Conn_01x04_Female	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical	1	
Q1,Q2,Q4,Q5	2N5088	Package_TO_SOT_THT:TO-92_Inline	4	
R1,R16	39k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	2	
R2,R8,R17,R19	100k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	4	
R3,R9,R13	470k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	3	
R4,R10,R14	15k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	3	
R5,R11,R15	100	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	3	
R6	1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	1	
R7,R12	8.2k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	2	
R18	390k	Resistor_THT:R_Axial_DIN0207_L6.3mm	1	

		_D2.5mm_P10.16mm_Horizontal		
R20	10k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	1	
R21	2.2k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	1	
RV1-RV3	100k	Potentiometer_THT:Potentiometer_Oeg_PC16BU_Vertical	3	

Total line items: 19